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Sub-surface post remover

Abstract

A sub-surface pole removal frame having a main bar extending downwardly from a handle bar and fixedly attached to a rectangular solid plate and connected to a pivotable work lever connected to an attachment member, first and second main struts extending downward at an angle to a first and second horizontally extending foot on opposite sides of the base, that form two triangular openings, such that the pole remover stands upright in a stable configuration with both feet on the ground, the solid plate provides a mount to secure a lifting mechanism with a flexible synching loop or grapnel, and a wheel assembly mounted to a rear portion of the frame, wherein the mount on the base is wedge shaped, wherein the wheel assembly is fixedly attached with two poles connected to the rear portion of the solid plate, wherein the struts are rigid tubular metal.

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Background/Summary

CROSS REFERENCE TO RELATED APPLICATION (1) This application is a continuation application and takes priority from application Ser. No. 17/571,608 entitled “Sub-Surface Post Remover” filed Jan. 10, 2022 now U.S. Pat. No. 12,012,774 issued on Jun. 18, 2024.

TECHNICAL FIELD
(1) The present invention relates to pole removers. More specifically the present invention relates to a portable subsurface pole remover for removing fence posts, sign posts and the like. Preferably, the pole remover can be operated by an individual person. The terms “pole” and “post” include poles or like elongate members, with or without a foundation, such as a foundation formed of concrete, and the term “post” is used hereinafter for convenience only and is not intended to be limiting on the scope of the invention.
BACKGROUND OF THE INVENTION
(2) Pulling fence posts or poles out of the ground is more difficult than most people realize. Fence posts are often used to support, e.g., backyard fences. When such fences are installed, the bottoms of the fence posts are typically encased in concrete so that the posts do not fall over and can support rails or fence segments that extend between the posts. Some posts may be placed in ground that may harden over time. When a concrete foundation is poured and hardened, the foundation may end up being relatively large and heavy. Foundations weighing upwards of 200 lbs. are

difficult to remove. Several people may have to dig a large hole over several days just to remove a single post and its associated concrete foundation.

(3) To address these issues, Great Britain Patent Document No. GB2511420, incorporated herein by reference, discloses an apparatus for raising a post. The apparatus includes a support frame that has a transverse base portion of rectangular shape. An arm is provided with a guide post which connects with the post. A ram has a fluid operated piston/cylinder device having a piston slidably mounted in a cylinder. The cylinder is connectable to a fluid source for actuating and extending the piston to raise the arm thus raising the post. However, this arrangement of GB2511420 does not connect to the post in a secure manner and has no provision for lifting buried posts.

(4) In addition, U.S. Pat. No. 6,598,856 discloses a portable hydraulic stake puller 1. During operation, a lifting tube 25 is raised by a hydraulic cylinder 21 pressing against the resiliency of spring members 43, 43 operably retained in spring tubes 45, 45. During downward operation, spring members 43, 43 return lifting tube 25 to its starting position. This operation occurs when a control lever 7 is released, allowing a valve 17 to return from an open position to a neutral position. The arrangement of U.S. Pat. No. 6,598,856 is designed to remove tent stakes. Like GB2511420, U.S. Pat. No. 6,598,856 does not connect to a post in a secure manner and has no provision for lifting buried posts.

(5) With the above in mind there still exists a need in the art for a system that is designed to pull a buried post or pole from the ground with or without a concrete foundation attached. There is also a need for the system to remove the concrete if the post has rotted or broken off at ground level. Preferably, the system should also be able to remove small trees and bushes, root and all, and remove the chore of having to dig each post out of the ground by hand. In addition, preferably the system should be portable and light weight enough for one person to move it around and use it in the environment they need to work in.

SUMMARY OF THE INVENTION

(6) The present invention is directed to a sub-surface pole remover. The pole remover includes a portable frame that can be moved by a standalone individual. On the frame there is mounted a pump configured to pressurize a fluid. In addition, a fluid cylinder is also mounted on said frame. The cylinder is configured to receive the pressurized fluid from the pump. A piston is slidably mounted in the cylinder. Preferably the fluid is air or hydraulic fluid but an electrical lift mechanism may be used in place of the fluid cylinder.

(7) The pole remover may have different arrangements for grabbing a fence post. In one preferable arrangement a steel rope or the like is attached to the lifting piston of the cylinder, at one end of the rope, and wrapped around a post at the other end of the rope. In another preferred arrangement, a sub-surface grapnel is connected to said lifting piston. The grapnel has a plurality of lower arms that extend from a central circular support and are configured to grip a sub-surface pole. A cable connects the grapnel to the piston. The grapnel has a top circular support and the central circular support is connected to the top circular support by a plurality of flexible cables (e.g., steel rope pieces). The middle circular support is connected to a plurality of upper arms and a bottom circular support is connected to a plurality of lower arms in a pivotable manner, and the upper arms are pivotably connected to the lower arms. Each lower arm includes gripping teeth configured to engage a pole or foundation and is also configured to be hammered into the ground.

(8) A control lever (e.g., a switch) is configured to direct the pressurized fluid from the pump to the cylinder whereby the cylinder is actuated to cause the piston to extend and push an attachment member upward relative to the portable frame and remove the sub-surface pole from the ground. Alternatively, the control lever may control an electric motor or lift mechanism.

(9) A method of using the pole remover includes gripping a sub-surface pole located below ground with the plurality of lower arms of the sub-surface grapnel. As noted above, the grapnel is connected to the lifting piston by the attachment member. The method further includes directing fluid from the pump to the cylinder to cause the lifting piston to extend and push the attachment

member upward relative to the portable frame and thereby remove the sub-surface pole from the ground. Preferably an individual using the grapnel starts by hammering the lower arms of the grapnel into the ground to engage the foundation with the gripping teeth of the lower arms.
(10) Additional objects, features and advantages of the present invention will become more readily apparent from the following detailed description of preferred embodiments when taken in conjunction with the drawings wherein like reference numerals refer to corresponding parts in the several views.

Description

BRIEF DESCRIPTION OF THE DRAWINGS
(1) The disclosure may be more completely understood in consideration of the following description of various illustrative embodiments in connection with the accompanying drawings.
(2) FIG. **1** is a perspective view of a sub-surface pole remover with a steel rope post gripper in accordance with a first preferred embodiment of the invention.
(3) FIG. **2** is a perspective view of the sub-surface pole remover of FIG. **1** pulling a fence post out of the ground.
(4) FIG. **3** is a perspective view of a sub-surface pole remover employing a grapnel in place of the steel rope post gripper of FIG. **1** in accordance with a second preferred embodiment of the invention pulling a fence post out of the ground.
(5) FIG. **4** is a side view of the grapnel of FIG. **3**.
(6) FIG. **5** is a close up view of a grapnel upper arm from the grapnel of FIG. **3**.
(7) FIG. **6** is a close up view of a grapnel lower arm with gripping teeth from the grapnel of FIG. **3**.
DETAILED DESCRIPTION OF THE INVENTION
(8) The following detailed description should be read with reference to the drawings in which similar elements in different drawings are numbered the same. The detailed description and the drawings, which are not necessarily to scale, depict illustrative embodiments and are not intended to limit the scope of the disclosure. Instead, the illustrative embodiments depicted are intended only as exemplary. Selected features of any illustrative embodiment may be incorporated into an additional embodiment unless clearly stated to the contrary. While the disclosure is amenable to various modifications and alternative forms, specifics thereof have been shown by way of example in the drawings and will be described in detail. It should be understood, however, that the intention is not to limit aspects of the disclosure to the particular illustrative embodiments described. On the contrary, the intention is to cover all modifications, equivalents, and alternatives falling within the spirit and scope of the disclosure.
(9) As used throughout this application, the singular forms “a”, “an” and “the” include plural forms unless the content clearly dictates otherwise. In addition, the term “or” is generally employed in its sense including “and/or” unless the content clearly dictates otherwise.
(10) Referring to FIGS. **1** and **2**, there is shown a perspective view of a first preferred embodiment of a sub-surface pole remover **10** that may be operated by a single user. Pole remover **10** has a handle bar **110** that is gripped by the user to move pole remover **10** from place to place. Preferably, handle bar **110** extends horizontally from a main bar **120**. Handle bar **110** is preferably provided with two grips, one for each hand of the user. The grips are not separately labeled. Handle bar **110** is preferably made of round steel pipe or a similar type of strong material but may be portable and light weight enough for one person to move it around and use it in the environment they need to work in.
(11) Main bar **120** extends downwardly from handle bar **110**. Main bar **120** is also preferably made of steel and is connected to a work lever **130**. Work lever **130** is mounted so as to be able to pivot relative to main bar **120**. Work lever **130** is also connected to an attachment member **140**. A hook or

pin **150**, connected to attachment member **140** provides a mounting for a first steel rope loop **160**. The first steel rope loop **160** is maintained in its shape by a first swage coupling **170**.

(12) Main bar **120** is also connected to a pair of main struts. A first main strut **180** extends downward at an angle to a first horizontally extending foot **190** to help support main bar **120**. A first support strut **195** extends between first main strut **180** and first foot **190** and forms a first support assembly **196**. Thus, main bar **120** is able to support work lever **130**. A second steel rope loop **200** is held in place by a second swage coupling **210** so that a steel rope **220** (which connects swages **170** and **210**) may be placed about an object (e.g., a sub-surface pole) above or below ground and apply a force to the object via work lever **130**. A second main strut **230** also extends between main bar **120** and a second horizontally extending foot **240** coupled with a second support strut **250** to form a second support assembly **255** with a triangular opening.

(13) Main struts **180**, **230**, support struts **195**, **250** and feet **190**, **240** are preferably all made of a strong material, such as 1 inch by 1 inch G carbon steel. Preferably support assemblies **196**, **255** extend symmetrically from the sides of main bar **120** such that pole remover **10** stands upright in a stable configuration with both feet **190**, **240** on the ground. A solid plate **257** extends between feet **190**, **240** and is also connected to main bar **120**, preferably by welding. Solid plate **257** provides a mount **260** with a pin to secure a lifting mechanism (described further below). A back wall **265** extends vertically along main bar **120**.

(14) A wheel assembly **267** is preferably mounted to the rear portion of solid plate **257**. Wheel assembly **267** includes two support legs, each leg including a horizontal support arm **268** and an angled support arm **269**. An axle **270** supports two wheels **280**. Each horizontal support arm **268** is connected to a horizontal axle holder **290** at one end and to a main bar support strut **300** at its other end. Each angled support arm **269** extends between plate **257** and its respective horizontal support arm **268** and provides additional reinforcement for horizontal axle holder **290**. Horizontal axle holder **290** preferably supports wheels **280**. Each wheel **280** has a rim that supports an approximately sixteen-inch non-inflatable polypropylene foam tire with a ¾ inch axel. Each wheel **280** is preferably rotatably supported on axle **270** mounted within horizontal axle support **290**.

(15) The lifting mechanism, mentioned above, includes a motor **310** that is preferably driven by compressed air, hydraulic fluid or electricity. A compressed air hose **320** extends from motor **310**, up along main bar **120** and to handle bar **110**. Alternatively, hose **320** could contain hydraulic fluid. A control lever in the form of a switch **370** is located in compressed air hose **320** near handle bar **110**. A connector **380** is provided for allowing compressed air or hydraulic fluid to be introduced from a compressor to hose **320**. Switch **370** may be actuated to allow compressed air or hydraulic fluid to travel to motor **310** or to prevent compressed air or hydraulic fluid from traveling to motor **310**. Preferably, a long ram jack **340** with an 8-ton rating is secured on mount **260**. Jack **340** is secured to the mount with a base lock including a pin configured to secure jack **340** to mount **260**. Jack **340** is connected at its upper end to attachment member **140** attached to work lever **130** by pin **350**. Work lever **130** is attached by a ½ inch carbon steel pin **360** to main bar **120** so that as jack **340** extends, work lever **130** pivots about pin **360**.

(16) Pole remover **10** has a relatively simple operation. A user grabs handle bar **110** and pulls back on main bar **120** to pivot and lift feet **190**, **240** off of ground **390** (labeled in FIG. **2**) such that the weight of pole remover **10** is supported by wheels **280**. The user then moves pole remover **10** to a fence post **400** and steel rope **220** is looped around post **400**, as shown in FIG. **2**. The user then actuates switch **370** to supply compressed air or hydraulic fluid to motor **310**. Motor **310** sends air or hydraulic fluid into jack **340**, causing jack **340** to extend toward work lever **130** as shown by an extending force arrow **430**. Work lever **130** pivots upward in respond to the extension of jack **340** as shown by a vertical force arrow **420**. Pole remover **10** remains stable due to feet **190**, **240** which are supported by ground **390**. As jack **340** extends, steel rope **220** tightens or cinches around fence post **400** and forces fence post **400** upward and out of ground **390** as shown by a vertical motion arrow **440**.

(17) In a second preferred embodiment, as shown in FIGS. **3-6**, pole remover **10** from the first embodiment, shown in FIGS. **1** and **2**, has been modified in two respects. First, axle support **290** is shown as mounted directly to plate **257**. It should be noted that wheel assembly **267** from the first preferred embodiment could be used in the second preferred embodiment and the wheel arrangement of the second embodiment could be used in the first embodiment.
(18) The second change is that steel rope **220** in the first embodiment has been replaced with a sub-surface grapnel **500**. The top of grapnel **500** is attached to work lever **130** by a connector **510** to a top circular support **530**. Connector **510** is connected to first steel rope loop **160** that is maintained in its oval shape **520** by a first swage coupling **170**. Grapnel **500** also includes a middle circular support **540** and a bottom circular support **550**. Bottom circular support **550** is in the shape of a circle and supports free moving lower arms **560**. Four lower arms **560** are shown in the drawings but three to six lower arms **560** are all considered preferable and six lower arms **560** is considered most preferable.
(19) Lower arms **560** are spaced evenly around bottom circular support **550**. As best shown in FIG. **6**, lower arms **560** each include a central circular shaped opening **585** to fit around bottom circular support **550**. Lower arms **560** are mounted for pivotal motion so as to be engaged with a foundation **562** of a fence post **564**, as shown in FIG. **4**. As shown in FIG. **6**, each lower arm **560** has a top lever arm **576** that extends from central circular opening **585** and terminates in a ring shape formed with a top circular opening **586** and a lower lever arm **578** that has a set of gripping teeth **580** extending to a tip **590**.
(20) A set of upper arms **570** is provided with each upper arm having a top circular opening **572** and a lower circular opening **574**, as best seen in FIG. **5**. Middle circular support **540** pivotably supports top circular opening **572** of each upper arm **570** in a pivotable manner. Lower circular opening **574** of each upper arm **570** is connected to top circular **586** opening of a respective lower arm **560** to form a pivoting joint.
(21) Top circular support **530** is connected to middle circular support **540** with a series of steel rope pieces **581** (flexible cables). Preferably, there are three to six steel rope pieces **581** and, most preferably, four steel rope pieces **581** are employed. The other parts of fence post puller **10** are preferably all made of 1½ inch schedule **40** carbon steel. The size of bottom circular support **550**, middle circular support **540** and top circular support **530** may vary. Preferably top circular support **530** has a 7-inch diameter, middle circular support **540** has a 20-inch diameter, and bottom circular support has an 18-inch diameter, but the relative sizes can be varied.
(22) Fence post remover **10** is designed to provide easy extraction of fence post **400** or fence post **564** and concrete foundation **562** surrounding post **564** that is situated below ground **390** as described below. Initially, fence post remover **10** is partially disassembled. Bottom circular support **550** and its associated free moving lower arms **560** are placed above ground **390** around fence post **564** that is considered a target fence post. Lower arms **560** are positioned around target fence post **564**. Lower arms **560** are then adjusted based on the size and shape of target fence post **564** and its associated foundation **562**. Lower arms **560** are then hammered into ground **390** around target fence post **564**. Middle circular support **540** and its associated set of upper arms **570** are then positioned over bottom circular support **550**. Lower circular opening **574** of each upper arm **570** is connected to top circular opening **586** of a respective lower arm **560** to form a series of pivoting joints. Top circular support **530** is then connected to middle circular support **540** with a series of steel rope pieces **581**. Top circular support **530** is also connected to attachment member **140**. These steps can be combined such that portions of pole remover **10** are pre-assembled before pole remover **10** is placed near fence post **564**. Lower arms **560** are then hammered into ground **390** around target fence post **564** and in either case the operation then proceeds in the same manner as described below.
(23) By actuating switch **370**, air or hydraulic fluid is sent through compressed air hose **320** to motor **310**. Motor **310** provides compressed air or hydraulic fluid to ram jack **340** to raise work

lever **130** and attachment member **140**. The jack **340** could also use an electrical system to provide the upward force. As an upward force is applied to top circular support **530** an upward force is applied to steel rope pieces **581**, this force also applies an upward force on middle circular support **540**. Lower lever arms **578** of lower arms **560** move inwardly toward each other when top lever arms **576** of lower arms **560** are pulled upwardly and outwardly. Each lower arm **560** pivots about bottom circular support **550** with bottom circular support **550** acting as a fulcrum to apply a lateral force on fence post **564** while also pulling fence post **564** upward. In a preferred embodiment, pressures up to 18,500 lbs. of force have been applied on fence post puller **10** in operation. Preferred embodiments of the invention may also be used for transport of other tools and articles on job site.

(24) Based on the above, it should be readily apparent that the present invention provides for an easy way for a user to remove a fence post. Even fence posts that are secured in place with a concrete foundation and have a portion of the fence post broken, can be easily removed. Although described with reference to preferred embodiments of the invention, it should be readily understood that various changes and/or modifications can be made to the invention without departing from the spirit thereof. For example, any of the features of one embodiment could be incorporated into any other embodiment.

Claims

1. A sub-surface pole and pole foundation removal frame comprising: a portable frame having a solid rigid rectangular plate comprising a first proximal and second distal opposing long sides and opposing short sides and with two forward extended legs from the base along the opposing short sides; a vertical arm fixedly attached and extended from the middle of the proximal long side of the base with two diagonal braces connecting the arm to the outer end of the base at the two opposing short sides; two diagonal arms fixedly attached to the approximate middle of the post at one end and connected at each of their distal ends to each of the forward extended legs; a pivoting arm extended from the post connected to a lifting piston; a flexible cinching loop or a grapnel operably connected to the pivoting arm; the lifting piston slidably mounted in a cylinder having a top and bottom and operably connected to the pivoting arm at the top of the cylinder; a pair of support posts connecting the diagonal arms at each respective middle portions of the diagonal arms to the distal side of the base where the base meets with forward extended legs; and an attachment on the base to which the bottom of the cylinder is positioned.
2. The sub-surface pole and pole foundation removal frame of claim 1, further comprising an attachment to the piston.
3. The sub-surface pole and pole foundation removal frame of claim 1, wherein the base is formed of plate metal.
4. The sub-surface pole and pole foundation removal frame of claim 1, wherein the pivoting arm is comprised of two parallel extensions attached to the vertical arm.
5. The sub-surface pole and pole foundation removal frame of claim 1, wherein the mount is wedge shaped to meet the bottom of the cylinder.
6. The sub-surface pole and pole foundation remover of claim 1, wherein the frame has a wheel assembly attached to the two diagonal braces connecting the arm to the outer end of the base at the two opposing short sides.
7. A sub-surface pole and pole foundation removal frame comprising: a main bar extending downwardly from a handle bar and fixedly attached to a rectangular solid plate; the main bar connected to pivotable work lever connected to an attachment member; a cinching loop or a grapnel operably connected to the pivotable work lever; a first main strut extending downward at an angle to a first horizontally extending foot on one side of the base to support the main bar; a first support strut extends between first main strut and the first foot and forms a first support assembly

with a triangular opening; a second main strut extending downward at an angle to a second horizontally extending foot on a second side of the base to support the main bar coupled with a second support strut to form a second support assembly with a triangular opening; the two support assemblies extend symmetrically from the sides of main bar such that pole remover stands upright in a stable configuration with both feet on the ground; the solid plate provides a mount to secure a lifting mechanism; a back wall extends vertically along main bar at the base; and a wheel assembly mounted to a rear portion of the frame.

8. The sub-surface pole and pole foundation removal frame of claim 7, wherein the mount on the base is wedge shaped.

9. The sub-surface pole and pole foundation removal frame of claim 7, wherein the wheel assembly is fixedly attached with two poles connected to the rear portion of the solid plate.

10. The sub-surface pole and pole foundation removal frame of claim 7, wherein the struts are rigid tubular metal.

11. The sub-surface pole and pole foundation removal frame of claim 7 further comprising rubber tires on the wheel assembly.

12. The sub-surface pole and pole foundation removal frame of claim 7 further comprising a pin for pivotable movement of the pivotable work lever.

13. The sub-surface pole and pole foundation removal frame of claim 7 further comprising directing hydraulic fluid to the cylinder.

14. The sub-surface pole and pole foundation removal frame of claim 7 wherein the wheel assembly comprises an axle connecting two wheels and the axle is fixedly attached to the plate.

15. The sub-surface pole and pole foundation removal frame of claim 7 wherein the attachment member is connecting to a wooden post.